

Task	Prompt Template	Prompt found by AUTOPROMPT	Label Tokens
Sentiment Analysis	{sentence} [T]...[T] [P].	unflinchingly bleak and desperate Writing academicswhere overseas will appear [MASK].	pos : partnership, extraordinary, ##bla neg : worse, persisted, unconstitutional
NLI	{prem}[P][T]...[T]{hyp}	Two dogs are wrestling and hugging [MASK] concretepathic workplace There is no dog wrestling and hugging	con : Nobody, nobody, nor ent : ##found, ##ways, Agency neu : ##ponents, ##lary, ##uated
Fact Retrieval	<i>X plays Y music</i> {sub}[T]...[T][P].	Hall Overton fireplacemade antique son alto [MASK].	
Relation Extraction	<i>X is a Y by profession</i> {sent}{sub}[T]...[T][P].	Leonard Wood (born February 4, 1942) is a former Canadian politician. Leonard Wood gymnasium brotherdicative himself another [MASK].	

Table 3: **Example Prompts** by AUTOPROMPT for each task. On the left, we show the prompt template, which combines the input, a number of trigger tokens [T], and a prediction token [P]. For classification tasks (sentiment analysis and NLI), we make predictions by summing the model’s probability for a number of automatically selected label tokens. For fact retrieval and relation extraction, we take the most likely token predicted by the model.

We use AUTOPROMPT with 5 or 7 tokens, and select the search parameters using the T-REx development set. We prevent proper nouns and tokens that appear as gold objects in the training data from being selected as trigger tokens. This is done to prevent AUTOPROMPT from “cheating” by embedding common answers inside the prompt. To evaluate, we observe the rank of the true object in label token distribution of the MLM, and use standard ranking metrics: mean reciprocal rank (MRR), precision-at-1 (P@1), and precision-at-10 (P@10).

Results Table 4 shows the performance of MLMs with different prompting methods, and we show qualitative examples in Table 3 and in Appendix C. Prompts generated using AUTOPROMPT can extract factual knowledge from BERT more effectively than their manual and mined counterparts: we improve P@1 by up to 12 points. Moreover, despite AUTOPROMPT using only one prompt per relation, it still outperforms LPAQA’s ensemble method (which averages predictions for up to 30 prompts) by approximately 4 points. Using 7 trigger tokens achieves slightly higher scores than 5 trigger tokens, although the difference is not substantial. This indicates that our approach is stable to the choice of trigger length, which is consistent with our sentiment analysis results. Overall, these results show that AUTOPROMPT can retrieve facts more effectively than past prompting methods, thus demonstrating that BERT contains more factual knowledge than previously estimated.

Relation Breakdown We also provide a detailed breakdown of the prompts found by Petroni et al. (2019) and AUTOPROMPT, and their associated accuracies in Appendix C, Table 7. Manual prompts are competitive when the prompt is *easy* to specify, e.g., the prompt “*was born in*” for the PLACE OF BIRTH relation. On the other hand, AUTOPROMPT performs especially well for relations that are difficult to specify in a natural language prompt. For example, Petroni et al. (2019)’s prompt for the POSITION PLAYED ON TEAM relation is “{sub} plays in [MASK] position”, which is not as specific as the relation requires. Although the prompt from AUTOPROMPT is not grammatical (“{sub} ediatric striker ice baseman defensive {obj}”), it does contain tokens that are directly related to sports.

BERT outperforms RoBERTa We finally directly compare BERT and RoBERTa. To do so, we subsample the LAMA test set to consist of examples where the object is a single token for both BERT and RoBERTa (*Original-RoBERTa*).⁴ BERT actually slightly outperforms RoBERTa, and we find that the prompts generated for RoBERTa tend to contain more irrelevant words (see Appendix C, Table 7). For example, the prompt generated by RoBERTa for the PLAYS INSTRUMENT relation contains words such as “Trump” and symbols such as “,” “()” for the POSITION PLAYED ON TEAM relation. It is surprising that RoBERTa does not

⁴The original dataset consists of examples where the object is a single token for BERT.

Prompt Type	Original			T-REx			Model	MRR	P@10	P@1
	MRR	P@10	P@1	MRR	P@10	P@1				
LAMA	40.27	59.49	31.10	35.79	54.29	26.38	BERT	55.22	74.01	45.23
LPAQA (Top1)	43.57	62.03	34.10	39.86	57.27	31.16	RoBERTa	49.90	68.34	40.01
AUTOPROMPT 5 Tokens	53.06	72.17	42.94	54.42	70.80	45.40				
AUTOPROMPT 7 Tokens	53.89	73.93	43.34	54.89	72.02	45.57				

Table 4: **Factual Retrieval:** On the left, we evaluate BERT on fact retrieval using the *Original* LAMA dataset from Petroni et al. (2019). For all three metrics (mean reciprocal rank, mean precision-at-10 (P@10), and mean precision-at-1(P@1)), AUTOPROMPT significantly outperforms past prompting methods. We also report results on a *T-REx* version of the data (see text for details). On the right, we compare BERT versus RoBERTa on a subset of the LAMA data using AUTOPROMPT with 5 tokens.

perform better than BERT, and it is worthy of investigating this further in future work. Additionally, recall that prompting is a *lower bound* on a model’s knowledge: the lower relative performance does not mean that the model actually knows less.

6 Relation Extraction

Apart from evaluating whether MLMs *know* facts, it is also important to evaluate whether they can *extract knowledge* from text. In this section, we use the task of relation extraction (RE)—to identify how entities are related in a given sentence—an important task in information extraction. We create RE prompts in a similar fashion as fact retrieval: for a given triple (subj,rel,obj) and sentence that expresses this relation, we construct a prompt as “{sent}{sub}[T]...[T][P].”, where the trigger tokens are specific to the relation, and label token is the correct object obj (see Table 3 for an example).

Setup We use the T-Rex dataset for RE because each T-REx fact comes with context sentences that mention the subject and object surface forms. We compare AUTOPROMPT to LAMA and LPAQA (their prompts are still useful here), as well as a recent supervised relation extraction model (Sorokin and Gurevych, 2017) that was also used by Petroni et al. (2019). To make the evaluation fair for the supervised RE model, we modify the standard RE evaluation. We give the model credit as long as it does not predict a different relation for the subject and object, i.e. we ignore the “no relation” prediction and all other relations. We also drop all sentences from evaluation for which the model’s named entity extractor failed to identify the subject and the object as entities. See Appendix B for further details. For the evaluation of all systems, we treat a prediction as correct if it is either the canonical version of the object (e.g., “USA”) or the

rendered surface form (e.g., “American”) for *any* of the context sentences in a given triple.

Results Table 5 shows the results for BERT and RoBERTa. MLMs can extract relational information *more effectively* than the supervised RE model, providing up to a 33% increase on the task when using AUTOPROMPT. RoBERTa also outperforms the supervised RE model, although it is worse than BERT (likely for similar reasons as we outline in Section 5). For both BERT and RoBERTa, we notice that the trigger tokens consist of words related to their corresponding relations (see Appendix D, Table 8 for full list), e.g. RoBERTa selects “*defy trademarks of namesake manufacturer*” for relation MANUFACTURER/PRODUCER OF PRODUCT.

Perturbed Sentence Evaluation A possible explanation for the strong results of MLMs in the RE setting is that they may *already* know many of the relations. Thus, they may directly predict the objects instead of *extracting* them. To separate this effect, we synthetically perturb the relation extraction dataset by replacing each object in the test data with a random other object and making the same change to the prompt. For example, “*Ryo Kase (born November 9, 1974 in ~~Yokohama~~ → Yorkshire) is a Japanese actor*” where Ryo Kase is the subject, Yokohama is the original object, and Yorkshire is the new object. We regenerate the prompts using the perturbed version of the data.

The accuracy of the RE model does not change significantly on the perturbed data (Table 5), however, the accuracy of the MLMs decreases significantly. This indicates that a significant portion of MLM accuracy comes from background information rather than relation extraction. Nevertheless, our prompts for BERT outperform their LAMA and LPAQA counterparts, which provides further evidence that AUTOPROMPT produces better probes.

Model	Original	Perturbed
Supervised RE LSTM	57.95	58.81
BERT (LAMA)	69.06	28.02
BERT (LPAQA)	76.55	30.79
BERT (AUTOPROMPT)	90.73	56.43
RoBERTa (AUTOPROMPT)	60.33	28.95

Table 5: **Relation Extraction:** We use prompts to test pretrained MLMs on relation extraction. Compared to a state-of-the-art LSTM model from 2017, MLMs have higher mean precision-at-1 (P@1), especially when using prompts from AUTOPROMPT. We also test models on sentences that have been edited to contain incorrect facts. The accuracy of MLMs drops significantly on these sentences, indicating that their high performance stems from their factual knowledge.

7 Discussion

Prompting as an Alternative to Finetuning

The goal of prompting a language model is to probe the knowledge that the model acquired from pre-training. Nevertheless, prompting has some practical advantages over finetuning for solving real-world tasks. First, as shown in Section 3, prompts generated using AUTOPROMPT can achieve higher accuracy than finetuning in the *low-data regime*. Moreover, prompting has advantages over finetuning when trying to solve *many different tasks* (e.g., the many users of the OpenAI GPT-3 API (Brown et al., 2020)). In particular, finetuning requires storing large language model checkpoints for each individual task, and drastically increases system cost and complexity because it requires deploying many different models at the same time. Prompting alleviates both of these issues. Only prompts are stored for each individual task, while the same pretrained model is used across all of the tasks.

Limitations of Prompting There are certain phenomena that are difficult to elicit from pretrained language models via prompts. In our preliminary evaluation on datasets such as QQP (Iyer et al., 2017) and RTE (Dagan et al., 2005), prompts generated manually and with AUTOPROMPT did not perform considerably better than chance. However, we cannot conclude that BERT does not know paraphrasing or entailment from these results. In general, different probing methods have different tasks and phenomena they are suitable for: AUTOPROMPT makes *prompt-based probes* more generally applicable, but, it still remains just one tool in the toolbox of the interpretability researcher.

Limitations of AUTOPROMPT One downside of AUTOPROMPT is that it requires labeled training data. Although this is also required for other probing techniques (e.g., linear probing classifiers), manual prompts rely on domain/language insights instead of labeled data. Compared to human-designed prompts, AUTOPROMPT generated prompts lack interpretability, which is similar to other probing techniques, such as linear probing classifiers. Another limitation of AUTOPROMPT is that it can sometimes struggle when the training data is highly imbalanced. For example, in Sections 4 and 5 we show that the prompts often just increase the likelihood of the majority label. Re-balancing the training data can help to mitigate this problem. Finally, due to the greedy search over the large discrete space of phrases, AUTOPROMPT is sometimes brittle; we leave more effective crafting techniques for future directions.

8 Conclusion

In this paper, we introduce AUTOPROMPT, an approach to develop automatically-constructed prompts that elicit knowledge from pretrained MLMs for a variety of tasks. We show that these prompts outperform manual prompts while requiring less human effort. Furthermore, the results for sentiment analysis and textual entailment suggest that, in some data-scarce settings, it may be more effective to *prompt* language models than to finetune them for the task. Although we focus only on masked language models in this paper, our method can be trivially extended to standard language models, and thus maybe useful for constructing inputs for models like GPT-3 (Brown et al., 2020). Source code and datasets to reproduce the results in this paper is available at <http://ucinlp.github.io/autoprompt>.

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